

Defects & Bonds in Solid

Defects in Solids:

The term Defect denotes a state of the lattice in which the atomic arrangement has departed from regularity. There are different kinds of defects;

- i. Point defects
- ii. Line defects
- iii. Plane defects
- iv. Electronic Imperfections
- v. Excitation states of crystals
- vi. Transient imperfections

The descriptions of defects are as follows:

1. **Point defects:** If the deviation from regularity is confined to a very small region of only about a few lattice constants, it is called a point defect. It is of four types:
 - i) Interstitial atom
 - ii) Schottky defect or vacancy
 - iii) Frenkel defect
 - iv) Substitutional impurity atom
2. **Line defects:** The example of this defect is dislocation. It is a region of strain in the crystal in which, part of the lattice has undergone a shearing strain equal to one lattice vector called a Burgers vector. It is of two types.
3. **Plane defects:** These defects have an extension in area and are confined to a small region. There are three types of plane defects:
 - i) Lineage boundary
 - ii) Grain boundary
 - iii) Stacking boundary
4. **Electronic Imperfections:** Errors in charge distribution or energy in solids are called electronics defects. These defects are responsible for important electrical and magnetic properties.
5. **Excitation states of crystals:** These are imperfections as these cause deviations from perfect crystal symmetry. Some examples are
 - i) Phonons and magnons
 - ii) Conduction electrons and holes
 - iii) Excitons
 - iv) Quantized plasma waves
6. **Transient imperfections:** These imperfections are introduced into the crystal from external sources, i.e.
 - i) Photons
 - ii) High energy charged particles like, electrons, protons, mesons and ions. High energy unchanged particles such as neutron and neutral

Crystal binding:

Crystal binding is formed by the attractive electrostatic interaction between the negative charges of the electrons and the positive charges of the nuclei which is entirely responsible for the cohesion of solids. Magnetic forces have only a weak effect on cohesion and gravitational forces are negligible.

The strength of crystal bond is measured by the energy required to dissociate a solid into isolated atoms, molecules or ions as appropriate and the energy is generally known as cohesive energy.

Different types of crystal bindings:

1. **Ionic Bond:** Ionic crystals are made up of positive and negative ions. The ionic bond results from the electrostatic interaction of oppositely charged ions.
Example: NaCl, CsCl, etc.
2. **Co-valent Bond:** The co-valent bond is the classical electron pair or homopolar bond of chemistry, particularly of organic chemistry. The cohesive forces in co-valent crystals arise from the sharing of electrons by adjacent atom. Each atom that participates in a co-valent bond contributes an electron to the bond.
Example: Diamond, Silicon, Germanium, etc.
3. **Metallic Bond:** The valence electrons of metal atoms are weakly bound. When such atoms interact to become a crystal, their valence electrons form a 'gas' of electron that move with relative freedom through the resulting assembly of metal ions. The characteristic feature of the metallic binding is the lowering of the energy of the valence electrons in the metal as compared with the free atom.
Example: Cu, Ag, Au, etc.
4. **Van der Waals Bond:** All atoms and molecules exhibit weak, short-range attractions for one another due to van der Waals forces. The bond formed by the van der Waals interaction is known as van der Waals bond. Van der Waals interaction are responsible for the condensation of gases into liquids and freezing of liquids into solids in the absence of ionic, co-valent or metallic bonding mechanisms.
Example: He, Ne, Ar, etc.
5. **Hydrogen Bond:** A neutral hydrogen atom has only one electron. It should form a co-valent bond with only one other atom. However, under certain conditions an atom of hydrogen is attracted by rather strong forces to two atoms, thus forming a hydrogen bond between them, with a bond energy of the order of 0.1 eV. A hydrogen bond occurs between certain molecules containing hydrogen atoms.
Example: HF_2^- , H_2O etc.

